Context-Aware ECG Diagnosis Using LLM-Guided Attention

Samrath Prakash, Kamakshi Rautela, and Kapil Gupta
School of Computer Sciences, UPES
Dehradun-248007, India
{samrath.14606@stu.upes.ac.in, kamakshi.rautela@ddn.upes.ac.in, kapil.gupta@ddn.upes.ac.in}

***Abstract*—Electrocardiogram (ECG) interpretation plays a critical role in early detection of cardiovascular diseases, such as myocardial infarction (MI), arrhythmias, and conduction disorders, due to its non-invasive, economical, and readily available nature. While recent developments in deep learning (DL) have improved automated ECG analysis, most of these approaches have static attention mechanisms that do not adapt to individual patients' clinical settings. In this work, we aim to overcome these challenges by proposing a safe and interpretable context-aware ECG diagnosis model with large language model (LLM) guided dynamic attention. Our proposed model enables contextual reasoning by integrating a CNN-LSTM model with an LLM-driven dynamic attention controller, which dynamically adjusts lead-wise and temporal attention in the ECG model. By separating the LLM from direct ECG signal processing, the framework guarantees safe operation and incorporates clinically relevant reasoning. The architecture is validated on the PTB-XL dataset for discriminating between myocardial infarction and healthy patients. We compare its performance against baseline and static attention models using accuracy, area under the receiver operating characteristic (AUROC) curve, precision, recall, and F1-score. The results show that the dynamic attention model guided by the LLM performs better and is more robust in terms of classification compared to the traditional baseline and static attention models. Moreover, the dynamic attention process focuses on relevant leads, improving model interpretability and trust. Our approach offers a safer and more flexible approach to intelligent ECG interpretation by bridging the static deep learning-based perception and context-based reasoning in clinical practice.**

***Index Terms*—ECG Analysis, Large Language Model (LLM), Static Attention, Dynamic Attention.**

I. Introduction

Heart disease is a major cause of death worldwide. Electrocardiograms (ECG) are commonly used for the diagnosis and management of cardiovascular diseases in clinical and emergent scenarios. But interpreting ECGs is time-consuming, requires expertise, and can be prone to variability. Over the past few years, deep learning (DL) approaches, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers, have shown great promise for classifying various ECG abnormalities, such as arrhythmia and myocardial infarction (heart attack). Moreover, public datasets such as PTB-XL have facilitated progress in this field by providing standardised evaluation and reproducibility [23]. In recent years, there has been a growing trend towards the use of Explainable Artificial Intelligence (XAI) methods, such as SHAP and LIME, to tackle interpretability issues by offering deeper insights into model reasoning. For example, the SpectroNet-LSTM model combines DL with explainable AI to diagnose heart diseases from physiological signals, highlighting that improved interpretability not only leads to better understanding of the system, but also facilitates higher trust among physicians [22]. While interpretable versions of SpectroNet-LSTM architecture offer post-hoc reasoning explanations, the reasoning process is fixed and not explicitly tied to patient-specific contextual information. However, while most current models have fixed attention patterns across samples and hence are "patient-agnostic". However, human reasoning is context-specific; cardiologists selectively attend to various leads and temporal intervals of ECGs based on patient symptoms, clinical suspicion, and individual characteristics. Furthermore, with the increasing use of transformer-based models and Large Language Models (LLMs) in ECG interpretation, recent research has raised key concerns regarding explainability, safety, and hallucinations when LLMs are used for diagnosis directly [1]. LLMs have shown impressive reasoning, contextual reasoning, and multi-modal processing abilities, and progress in ECG modelling [16]. In healthcare, they have been used for multi-modal diagnosis, report generation, summarization of clinical text, and answering questions. Recent studies have also ventured into ECG language alignment and multi-modal LLMs to provide diagnostic and rationale explanations of ECGs. But the direct use of LLMs in diagnosis raises concerns over their clinical utility, safety and hallucination issues [26]. This research is driven by the idea that LLMs do not need to be used as main predictors for diagnosis to be used in medical AI systems. Rather, they excel in reasoning and decision-making. By using LLMs as attention controllers, rather than predictors, it's possible to direct the attention of DL models towards signal regions of clinical importance without sacrificing the reliability and integrity of signal-based diagnosis. The major contributions of this work are as follows:

- We develop a new dynamic context-aware ECG classification model to detect MI from normal samples by leveraging the high-level reasoning capabilities of LLMs to enable dynamic and patient-specific ECG analysis, in combination with DL.
- The proposed framework adopts a dynamic attention

mechanism, where the LLM serves as a dynamic attention controller, aggregating and synthesising the attention weights across time and leads according to the patient's metadata (age, gender, height, weight, and body mass index (BMI)) to generate attention maps.
- To overcome class imbalance, the proposed approach employs a transformer-based diffusion model, TransDiffECG, for generating clinically relevant synthetic ECG signals to increase data variability and model generalisability.
- The proposed framework achieves better accuracy, F1-score and AUROC metrics. The proposed approach improves interpretability by visualising lead-wise attention in accordance with well-established clinical thought processes, thus enhancing the transparency and trustworthiness of medical AI models.

II. Literature Review

ECG processing and interpretation have been the subject of research for decades because of their potential for the diagnosis of heart disease. The advent of DL and the availability of large, annotated datasets have greatly improved diagnostic accuracy. However, issues with clinical relevance, transferability, and interpretability remain. Initial DL-based methods for ECG analysis mainly involved CNNs applied directly to the ECG waveform. The release of the PTB-XL dataset provided a common standard for ECG interpretation, showing that CNN architectures can perform well in various cardiac classification tasks [23]. But follow-up research found that these approaches tend to overfit the training data, leading to poor performance across a range of clinical scenarios. To overcome these challenges, there has been a growing emphasis on enhancing model robustness and generalizability. Practical assessments have demonstrated that DL networks tend to overfit ECG data and perform poorly on out-of-distribution test data, highlighting the need for more dynamic and contextual modelling approaches [2]. While model architecture improvements have boosted classification performance, the problem of static feature extraction remains. Subsequently, attention mechanisms have been integrated to improve performance and interpretability. Techniques employing channel attention, multi-head attention, and hybrid CNN-LSTM models have shown better performance in the detection of arrhythmias and multi-label classification of ECGs [8]. While these approaches have improved performance, the attention weights learned remain fixed during inference and are globally optimized, making them less patient-specific. To enhance interpretability, explainable artificial intelligence (XAI) methods, such as saliency maps, Grad-CAM and SHAP, have also been extensively used in ECG analysis. These techniques offer post-hoc explanations, but do not impact the model's decision-making process or feature learning. As a result, there is a disconnect between the explanation and the prediction, limiting their effectiveness for clinical practice [17]. The transformers have built on the shortcomings of CNNs by incorporating self-attention mechanisms to model long-term dependencies in ECGs. These have demonstrated better results in the prediction of long-term and rhythm-level patterns. Recent reviews on transformer- and LLM-based ECG analysis point to the growing trend in applying large attention mechanisms for clinical ECG analysis. But they also highlight the potential risks of hallucination, lack of calibration and insufficient safety when LLMs are directly applied for diagnosis. Other recent studies have also investigated the use of LLMs for multimodal ECG analysis. Systems like ECG-Chat and ECG Semantic Integrator integrate ECG signals with textual medical records for generating reports and zero-shot diagnosis [27]. While these methods showcase the reasoning capability of LLMs, they often use LLMs as primary diagnostic tools, which could be potentially risky and unreliable for clinical use. A related approach is the LLM-Informed ECG Dual Attention Network (ECG-DAN) that uses 12-lead ECG signals to predict the risk of heart failure [6]. It proposes a lightweight dual-attention network, which includes a cross-lead attention mechanism to model the spatial relationships between different ECG leads and a temporal attention mechanism to learn the important segments of the ECG signals. An important aspect of this work was the use of an LLM in the pretraining stage, where BioClinicalBERT was used to embed semantics obtained from structured ECG reports in the PTB-XL dataset [15]. These embeddings are integrated with latent representations of the ECG using multimodal learning to enable the model to learn clinically relevant features. The current work builds on these developments to apply context-aware dynamic attention to the task of ECG diagnosis. Unlike previous approaches, the LLM is used to guide adaptive attention mechanisms based on patient metadata, without being exposed to training on the actual ECG signals or making diagnostic predictions. This approach guarantees greater clinical safety while facilitating more dynamic, explainable and context-sensitive ECG interpretation. The application of CNNs, long LSTMs, transformers, and hybrid attention networks have greatly improved ECG classification accuracy. However, a key issue remains: most models use fixed attention mechanisms, where attention weights are not responsive to the patient's clinical data, such as age, sex, risk factors, family history, or clinical suspicion of certain conditions. As a result, these models do not dynamically focus on different aspects of the ECG for each patient, which deviates from the contextual and adaptable reasoning used by real-world cardiologists. Various XAI techniques such as Grad-CAM, SHAP, and saliency maps have been extensively used to improve model interpretation. But these approaches are post-hoc and do not directly affect the model's reasoning process. This leaves a substantial gap between model understanding and explanation, which hampers their ability to enhance clinical trust and utility. Even in the most similar methods, which integrate LLMs in the pretraining phase to align features, the LLMs are used only for pretraining - not for reasoning during inference time. While these approaches enhance feature representation and interpretability, the attention mechanisms are fixed at the end of training, and can't adapt to dynamic, contextual inference during deployment.

The research compared the proposed method with the current state-of-the-art methods for ECG classification that were applied on the PTB-XL dataset to place it quantitatively. Conventional CNN models, attention-based models and more recent LLM integrated models are compared.

A. Methodology

1) Dataset Description: This study uses a publicly available large-scale ECG database, the PTB-XL dataset, as the source of data because of its volume, clinical significance, and annotations. The ECG signals were recorded over a period of nearly seven years (October 1989 to June 1996) with the equipment of Schiller AG. It contains 21,837 clinical 12-lead ECG recordings from 18,885 patients (52% men and 48% women). It covers a broad age distribution (0-95 years), with a median age of 62 years and an interquartile range of 22. The recordings are 10 seconds long [24]. Two cardiologists provided labels to the ECG signals, which are multiple diagnostic statements. There are 71 statements in total, which are divided into 44 diagnostic statements, 12 rhythm statements and 19 form statements, with some statements falling into multiple categories. Further, a hierarchical labeling is provided for diagnostic statements, which includes five coarse-grained superclasses and 24 fine-grained subclasses [21]. We select a binary classification task, myocardial infarction (MI) versus non-myocardial infarction (NORM), for this study. This task is dependent on both inter-lead and temporal features of the ECGs, making it ideal for assessing the performance of dynamic attention models. In this study, 500 Hz ECG signals are used as the input to our framework.

2) System Architecture and Design: The proposed system is an end-to-end intelligent ECG analysis framework for automated Myocardial Infarction (MI) detection. The architecture combines categorization, dynamic attention learning, temporal sequence modeling, deep feature extraction, and data preparation into a single pipeline. The system is divided into five layers:

- Data Acquisition Layer: The PTB-XL dataset contains 21,837 clinical 12-lead ECG recordings which are collected from 18,885 patients. Each ECG recording has a duration of 10 seconds with a sampling frequency of 500 Hz and an additional down sampled version of 100 Hz to facilitate computational efficiency during machine learning experiments. Since the raw ECG signal is a noisy electrical recording of the heart, it contains the real heart signal (P-QRS-T waves), slow drifts, electrical interferences, muscle noise and amplitude differences across leads it is important to remove these.
- Preprocessing Layer: The preprocessing pipeline goes through different phases where it removes different types of noise from the ECG signals.
- Trans-Diff Model: Due to high class imbalance between the Normal and MI with 78% classes as normal and 22% MI. To handle this class imbalance a transformer-based Diffusion model is implemented to handle the class imbalance between the Normal and Myocardial Infarction. The model of TransDiffECG integrates a diffusion-based generative framework with transformer-based and semantic conditioning modules. It takes a noisy ECG signal, a diffusion timestep, and a semantic segmentation map (indicating phases of P, QRS, T waves) as inputs. The core components include residual blocks with Transformer encoders/decoders to model global temporal dependencies, and Semantic ECG Batch Normalization (SEBN) blocks that incorporate phase-specific semantic information directly into feature normalization. This enables the model to generate physiologically accurate, controllable ECG signals by conditioning on detailed semantic masks and leveraging self-attention mechanisms for capturing long-range dependencies [14].

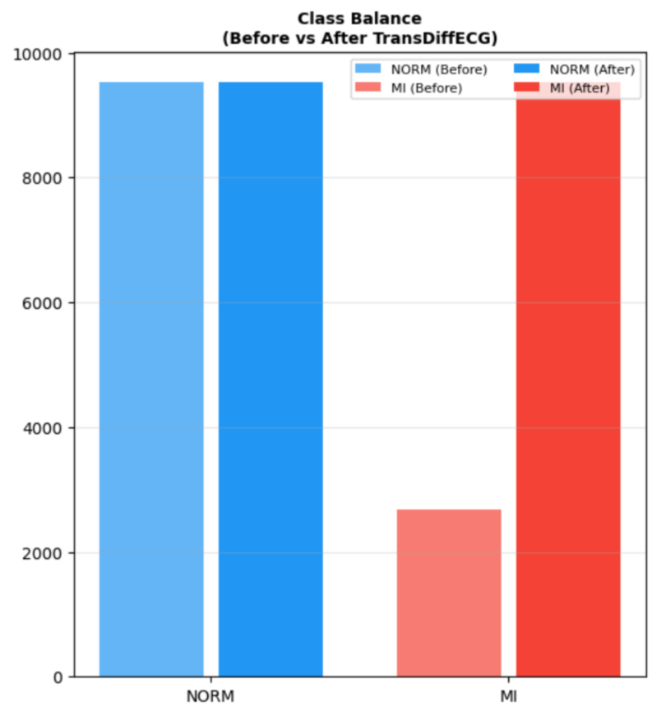

Fig. 1. Before and After TransDiff Augmentation

- **Slow vertical drifting of the signal baseline**: which moves up and down because of breathing, body movement and electrode impedance
- **Electrical Inference:** the electrical devices create 50 Hz sinusoidal interference using Notch Filter [4].
- **High Frequency Noise:** comes from muscle activity, sensor noise and electrode movement using low-pass filter.
- **Amplitude Difference between electrodes:** Normalizing the different ECG recordings having different amplitudes using Z-Score.

- Feature Extraction Layer:
Once the data is preprocessed it is first given to the 1D-CNN model for both feature extraction across the leads and for training the baseline CNN. Consists of 3

TABLE I
STATE-OF-THE-ART ECG CLASSIFICATION METHODS

Existing Models	**Dataset**	**Architecture**	**LLM Usage**	**Attention**	**Strength**	**Limitation**
Li et al. [11] 2017	PTB-XL	1D CNN	No	None	Strong local feature extraction	No temporal modeling, no interpretability
Rai et al. [20] 2022	PTB-XL	CNN + LSTM	No	None	Captures temporal dependencies	No attention mechanism
Kumari et al. [10] 2025	PTB-XL	CNN-BiLSTM	No	Static Attention	Improved feature learning	Fixed attention during inference
XAI-based Models [17] 2025	ECG Datasets	CNN + Hybrid	No	Post-hoc	Improved interpretability	No effect on model reasoning
Transformer Models [1] 2025	ECG Dataset	Transformer	No	Self-Attention	Global dependency modeling	Highly complex, lacks clinical context
ECG-Chat [26], [27] 2024-25	ECG + Text	Multimodal LLM	Yes	Implicit	Strong reasoning capability	Safety issues, hallucinations
ECG-DAN [5] 2025	PTB-XL	Dual Attention + LLM	Yes (Pretraining)	Static	Improved feature alignment	No dynamic inference-time attention
Kerdoudi et al. [9] 2026	PTB-XL	CNN + Multi-head Attention	No	Static	Captures long-range dependencies	No context adaptation
Proposed Model 2026	PTB-XL	CNN + LLM Dynamic Attention	**Yes (Inference)**	**Dynamic Attention**	Context-aware, interpretable	—

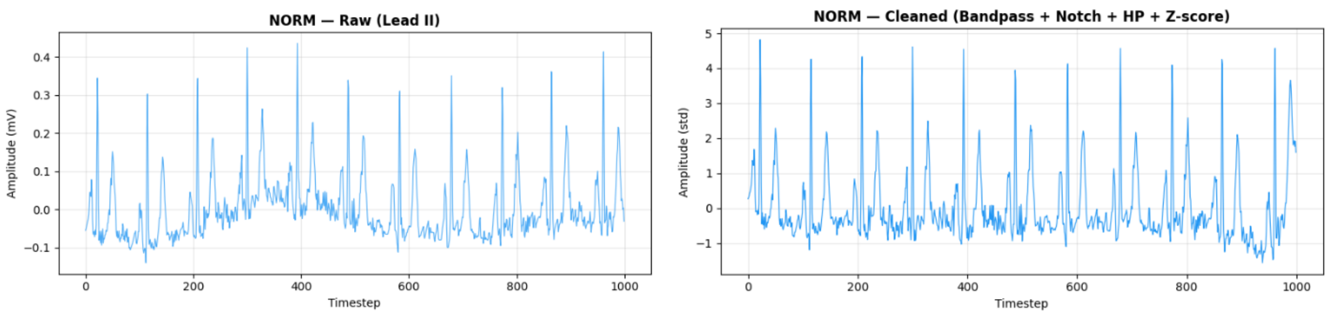

Fig. 2. Raw Normal ECG Signal for Lead II

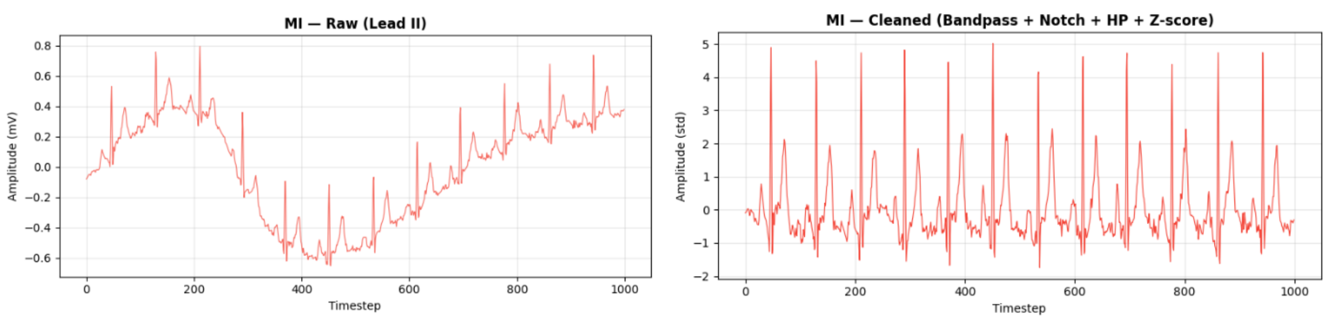

Fig. 3. Cleaned MI ECG Signal for Lead II

convolutional 1D Block with 32,64,128 feature maps, kernel size of 7,5,5,3 and pooling of 2 at different layers respectively. An Adaptive Average Pooling is applied to automatically calculate the kernel size and stride. At the last layer consist of a flatten layer of size (128, 2) [12]. Later the preprocessed data is given to the hybrid model of CNN + LSTM to first capture local morphological features and then feed the resulting sequence into a bi-directional LSTM to capture long-range temporal dependencies across the cardiac cycle, the bi-directional LSTM with a hidden state of 128 is used during the feature extraction phase. [19].

- Temporal and Attention Learning Layer:
  After the feature extraction layer the output is passed on the LSTM Layer which learns long-term temporal dependencies in ECG signals. The CNN layer which is responsible to captures local features, LSTM captures sequential dependencies over time since ECG is a time-series signal. The LSTM model remembers previous cardiac events while processing current time samples [13]. The attention layer is responsible for focusing on the most relevant ECG leads which enables the model to learn the most important weights because of which the model achieves higher test accuracies. In this research we implemented 3 different attention models.

  – **Model 1: 1D CNN baseline**
    The 1D CNN is a baseline model that has been used in large number of research and is one of the most preferred model in classification of ECG signals. The model extracts local temporal features

by stacking each of the convolutional layers [12]. Each of the Convolutional 1D Block consists of two convolutional layers with Batch Normalization after each layer, ReLU Activation function and a Max Pooling layer of size 2 and Dropout value of 0.2. The structure enables to model to learn local temporal patterns, maintain training stability and reduce overfitting.

TABLE II
1D CNN MODEL LAYER CONFIGURATION

Layer	Filters	Kernel Size	Stride
Conv1	32	7	1
Conv2	64	7	1
Conv3	128	7	1
Conv4	256	7	1

– **Model 2: 1D CNN with static attention**:
Using 1D CNN baseline model that captures QRS morphology, ST elevation/depression, T-wave abnormalities, waveform peaks and valleys [7]. A normalized ECG tensor with the following dimensions is sent to input layer (B,12,1000). Every lead is handled as a separate input channel, with this configuration the network can learn lead-wise morphological features.

TABLE III
CNN MODEL LAYER CONFIGURATION

Layer	Filters	Kernel Size	Stride
Conv1	32	7	1
Conv2	64	7	1
Conv3	128	7	1
Conv4	256	7	1

$$y(t) = \sum_{k=0}^{K-1} w_k \, x(t-k) + b \tag{1}$$

In each convolution layer a ReLU activation function is used which improves non-linearity and accelerates convergence. A max-pooling layer of pool size 2 is used to improve computational efficiency and prevents overfitting. In the dropout layer a dropout probability of 0.3 is used after each block which improves generalization. Multiple convolution blocks learn hierarchical ECG representations, after which static attention is applied, the static attention layer uses a softmax activation function to assign probability scores to each of the weights.These weights remain fixed after training and do not adapt dynamically based on the patient sample.

– **Model 3: CNN-BiLSTM baseline**
By capturing both spatial and temporal features of the ECG signals, the suggested model efficiently classifies ECG signals using a hybrid Convolutional Neural Network (CNN) and a Bi-directional Long Short Term (BiLSTM) architecture. By using several convolutional and pooling layers, The CNN component is first used to extract significant local features from the ECG waveform, such as P-wave, QRS complex and T-wave, these layers allow the model to concentrate on crucial morphological features by lowering the dimensions and improve feature representation. The Bi-LSTM is specifically used to learn the temporal connections in sequential data, receives the extracted features which are particularly important for detecting subtle and complex cardiac abnormalities. The Bi-LSTM processes forward and backward to capture long term dependencies and relationships between distant ECG points The following equations show the forward and backward implementation of BiLSTM.
Forward Pass:

$$\mathbf{A}_i = f_1 \left(\mathbf{W}_1 \mathbf{x}_i + \mathbf{W}_2 \mathbf{A}_{i-1} \right) \tag{2}$$

Backward Pass:

$$B_i = f_2 \left(\omega_3 x_i + \omega_4 B_{i+1} \right) \tag{3}$$

Output:

$$Y_i = f_3 \left(\omega_5 A_i + \omega_6 B_i \right) \tag{4}$$

TABLE IV
CNN LAYERS

Layer	Filters	Kernel Size
Conv1	32	7
Conv2	64	5
Conv3	128	3

– **Model 4: CNN-BiLSTM with static attention**:
This model integrates and extends the CNN baseline by integrating the temporal learning sequence. The CNN is responsible to capture the local morphological features which are then passed on to an LSTM network. This is important because cardiac abnormalities are temporal in nature. The same input (B,12,1000) is used with the following configuration. The LSTM layer the CNN output is reshaped

TABLE V
CNN LAYERS

Layer	Filters	Kernel Size
Conv1	32	7
Conv2	64	5
Conv3	128	3

and passed on to the Bi-directional LSTM. The configuration has a hidden size (128), layers (1), bidirectional (True), output size (256) because of forward, backward pass. The LSTM state update is given by equation:

$$h_t = f(x_t, h_{t-1}) \tag{5}$$

it captures sequential waveform dependencies. In order to improve the model's ability to focus on important features while ignoring unnecessary information, an attention mechanism is also implemented after the BiLSTM layer to give greater weights to the most relevant areas of the ECG signals. In order to classify the various heart states, the revised feature representations are then given as input to the fully connected layers. By utilizing the advantages of both deep learning approaches, the combined CNN-BiLSTM framework greatly improves the classification performance, leading to better accuracy, robustness and generalization of ECG signal processing [10].

– **Model 5: Proposed LLM Based Dynamic Attention**:
The proposed study expands on the static attention framework. The attention mechanism is dynamically created utilizing contextual information rather than set attention weights learned during training. This framework employs a Large Language Model as a high-level reasoning module that is fed structured patient context, like age, bmi, height, weight, gender of the patient details for the diagnostic task. The LLM creates structured attention guidance that indicates the relative significance of temporal areas and ECG leads based on this context [25]. In this study we used Llama 3.1 as our LLM model. A system prompt with the following "You are a certified cardiologist specializing in ECG interpretation and Myocardial Infarction detection. Given the patient's clinical data your task is to assign important weights to each of the 12 ECG leads for MI detection in that specific patient. Clinical reasoning guide for Inferior MI (RCA territory) most important leads are II, III,aVF, for Anterior MI (LAD territory) most important leads are V1,V2,V3,V4, for Lateral MI (LCX territory) most important leads for I, aVL, V5, V6. Elderly patient have higher risk for multi territory involvement. Higher BMI correlates to inferior and anterior MI" is then created that is used as a prompt template. Further an example output in JSON is given as "I": 0.07, "II": 0.12, "III": 0.10, "aVR": 0.05, "aVL": 0.06, "aVF": 0.11, "V1": 0.09, "V2": 0.10, "V3": 0.09, "V4": 0.08, "V5": 0.07, "V6": 0.06, the given system prompt is send to Llama 3.1 LLM model which has a temperature value of 0.1 to avoid randomness and generate deterministic and stable outputs, a top-p of 0.9 is used for nucleus sampling that selects tokens from the smallest probability covering 90% of the distribution. In order to prevent the LLM from generating excessively long output a num-predict parameter with value of 200 is used. The LLM with its existing knowledge of the corpus specifies the leads that the model should focus on. A trainable neural gating mechanism that learns lead-wise relevance ratings from the recovered ECG representations create the lead significance vector. Adaptive significance weights are generated by projecting the 12-dimensional lead representation through fully connected layers and normalizing it with a sigmoid activation function. Each ECG sample is dynamically adjusted, with the help of backpropagation these weights are learned during training. The CNN encoder transforms these instructions into numerical attention weights, which are then applied to the feature maps. The model's behavior is in line with clinical reasoning techniques because the dynamic attention mechanism allows the model to adjust its focus based on the clinical context unique to each patient, The LLM is only utilized as an attention controller and not for direct diagnosis. This ensures system safety and reduces the possibility of hallucinations associated with LLM-based medical predictions. Also a safety mechanism is also applied to the existing model if the model fails to generate dynamic attention it will automatically fall to static attention.
The following equation is adapted from [3]. The Dynamic Attention Gate is the most crucial aspect and novelty of this proposed model. The vector projected for the 12 lead is as follows 12$->$64$->$128 using the fully connected layers. By using the sigmoid activation function the gated feature becomes

$$X' = X \odot G(w) \tag{6}$$

this dynamically changes the attention per patient sample. The attention layer re-evaluate the temporal attention weights by using the equation:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{i=1}^{T} \exp(e_i)} \tag{7}$$

which focuses on diagnostically important intervals.
The complete system architecture is shown in Fig: 4

TABLE VI
CNN FEATURE EXTRACTOR

Layer	Filters	Kernel Size
Conv1	32	7
Conv2	64	5
Conv3	128	3

- Classification and Evaluation Layer:
The classification layer is the final stage of the proposed ECG diagnosis framework and is responsible for assigning each ECG signal to diagnose two categories Class 0: Normal (NORM), Class 1: Myocardial Infarction (MI). The main purpose of this layer is to convert the learned feature space into binary probability score that indicates the risk of myocardial infarction. The input for the classification layer differs depending on the architecture used. For the 1D CNN with static attention, the classifier is given an attention refined feature representation. Prior to categorization, the attention layer improves

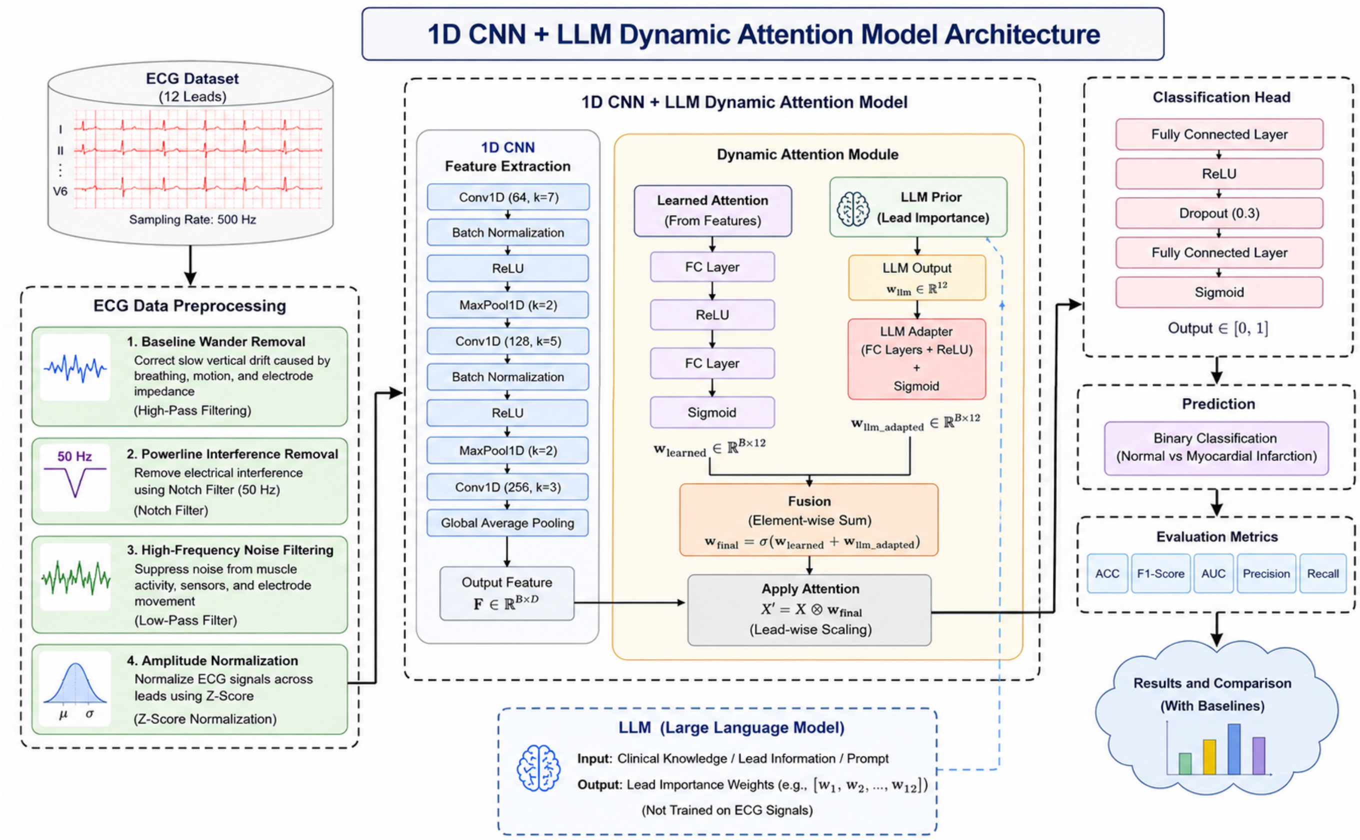

Fig. 4. 1D CNN + LLM Dynamic Attention Architecture

diagnostically significant channels. The CNN+BiLSTM with static attention classifier produces the attention-weighted temporal context vector which focuses on clinically significant time steps. The extracted feature vector are then passed through a fully connected (dense) layers. The layers perform linear transformation: z = Wx + b. The output logits were then passed through sigmoid activation function which are converted into probability scores. A probability (p) threshold value of 0.5 is used for final classification, The decision rule is based on

$$\hat{y} = \begin{cases} 1, & p \geq 0.5 \\ 0, & p < 0.5 \end{cases} \tag{8}$$

if p $\leq$ 0.5 the ECG is classified as Myocardial Infarction (MI), while p $<$ 0.5 the ECG is classified as Normal (NORM), this threshold value ensures a balanced decision boundary between the two classes. The output of the classifier is then optimized using a Binary Cross-Entropy Loss (BCE Loss) [18]. The equation is given by:

$$L = -\left[y \log(p) + (1 - y) \log(1 - p)\right] \tag{9}$$

B. Experimental Results

The experiments were conducted on the PTB-XL 12-lead ECG dataset for binary classification of Myocardial Infarction (MI) and Normal (NORM) classes. The dataset was split into 60% training, 20% validation, and 20% testing sets.

The models were trained on NVIDIA RTX 1650 using AdamW optimizer with a learning rate of 3e-4, batch size of 64, and trained for 50 epochs. Early stopping (patience = 4) and (CosineAnnealing) scheduler were used to improve convergence and prevent overfitting. Performance was evaluated using Accuracy, Precision, Recall, F1-score, and AUC shown in Table: VII.

In addition to it Training Loss as shown in Fig: 5, Training Accuracy shown in Fig: 6, Confusion Matrix shown in Fig: 7 and ROC-Curve shown in Fig: 8 are also validated to show the model's robustness.

The proposed dynamic attention model integrated with the 1D CNN backbone which extends the static attention demonstrates stable convergence and improved classification performance. The training curves indicate smooth optimization behavior with minimal overfitting. The confusion matrix shows strong discrimination between MI and NORM classes, and the ROC curve confirms high separability.

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Parameters	Accuracy	Precision	Recall	F1-score	AUC
1D-CNN	949,058	0.9633	0.9850	0.9412	0.9624	0.9921
1D-CNN + Static Attention	949,326	0.9590	0.9501	0.9680	0.9591	0.9894
1D-CNN + Dynamic Attention (LLM)	1,820,686	**0.9642**	0.9721	0.9554	**0.9635**	**0.9947**
CNN + Bi-LSTM	918,850	0.9625	0.9830	0.9412	0.9622	0.9915
CNN + Bi-LSTM + Static Attention	919,118	0.9565	0.9452	0.9690	0.9570	0.9940
CNN + Bi-LSTM + Dynamic Attention	3,934,030	0.9633	0.9783	0.9481	0.9631	0.9923

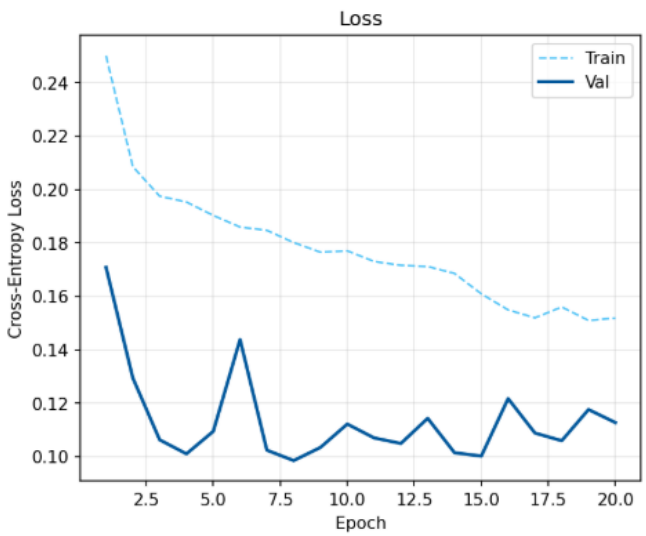

Fig. 5. Loss Curve

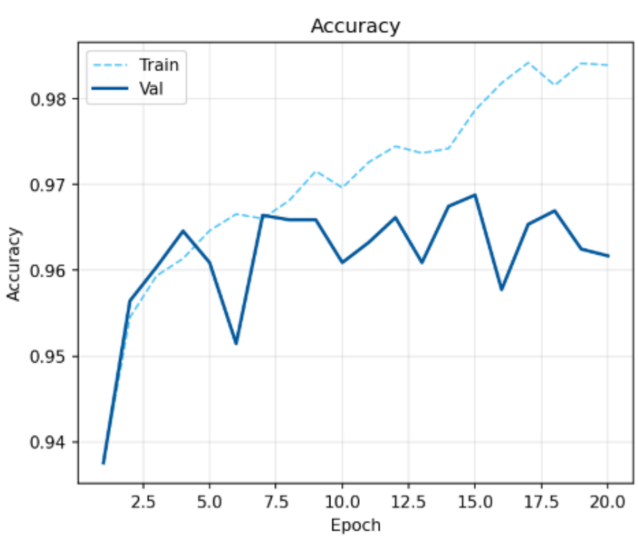

Fig. 6. Training Accuracy

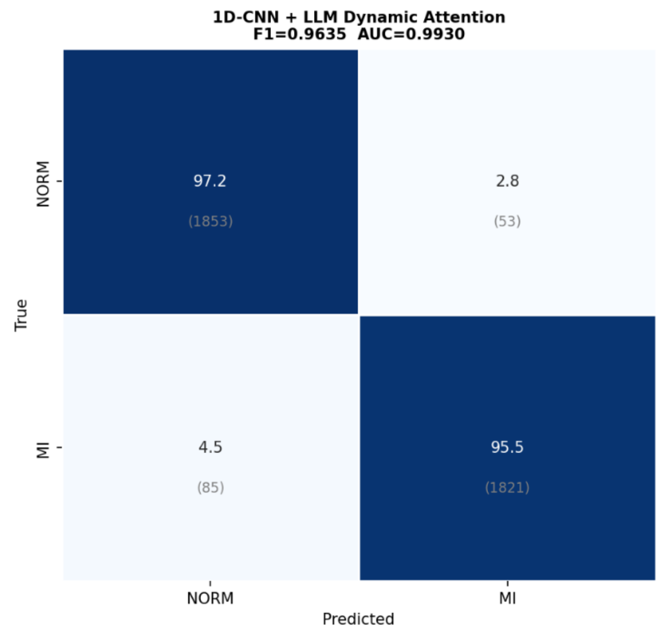

Fig. 7. Confusion Matrix

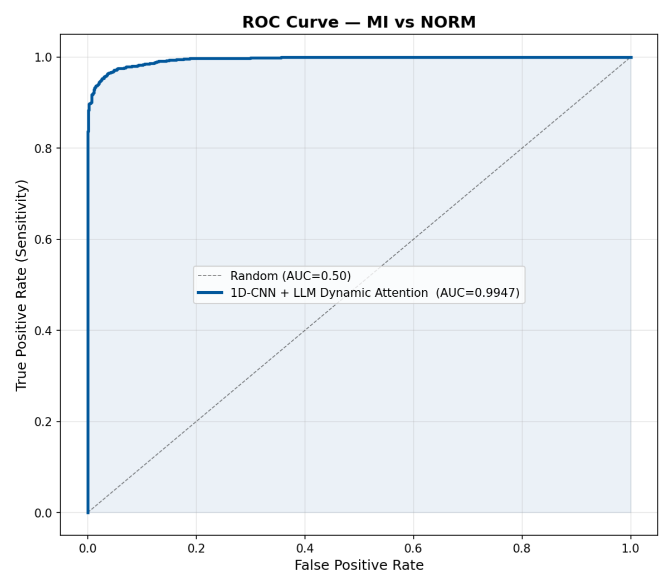

Fig. 8. ROC-Curve

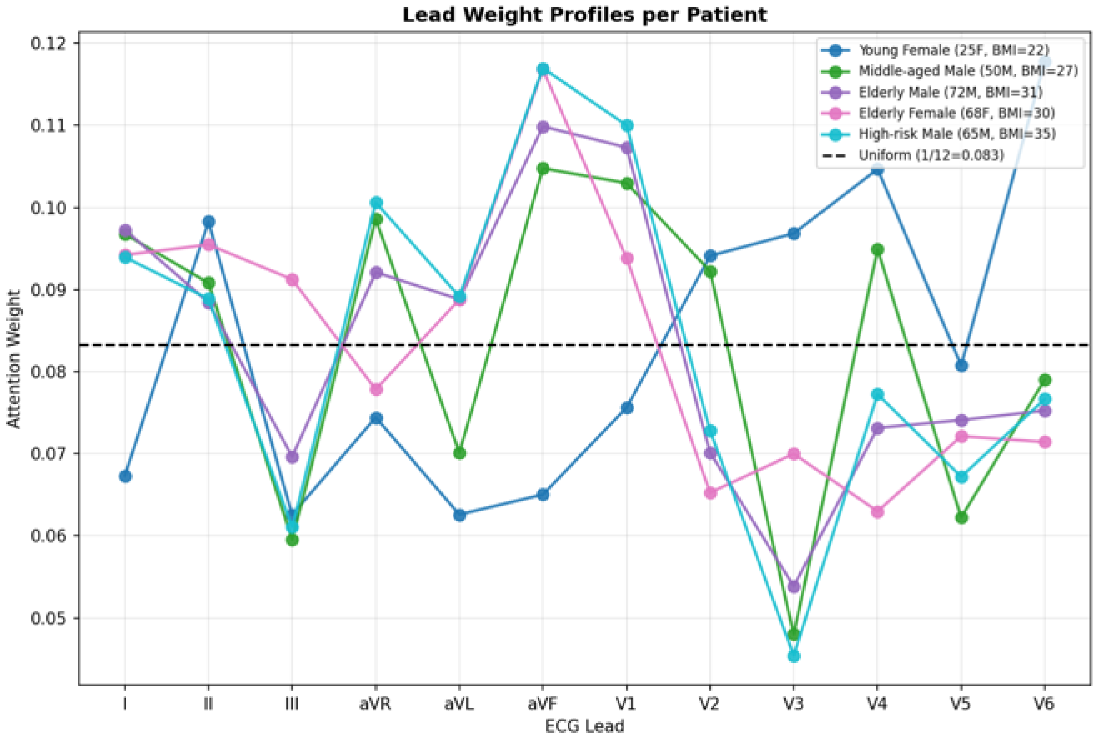

Fig. 9. Lead-wise attention visualization.

C. Comparative Performance Analysis The experimental results show, that although all models perform well in classification shown in Fig:10, there are significant variations in their capacity to maintain balance between precision and recall, which is essential for medical diagnosis.

The proposed 1D-CNN with LLM guided dynamic attention achieves strong performance due to its ability to capture local ECG features. The proposed model shows a significant improvement in Accuracy, F1-Score and AUC metrics without effecting much on the precision and recall, making it a strong model for our study. The lead attention visualization based on the patients profile shown in Fig: 9 demonstrates that the model dynamically focuses on clinically relevant ECG leads depending on patient context, validating the effectiveness of LLM-guided attention.

The results show that the proposed LLM-guided dynamic attention models achieve competitive or superior performance compared to baseline and static attention models. In particular, the dynamic attention mechanism improves F1-score and AUC without effecting much on Precision and Recall metrics.

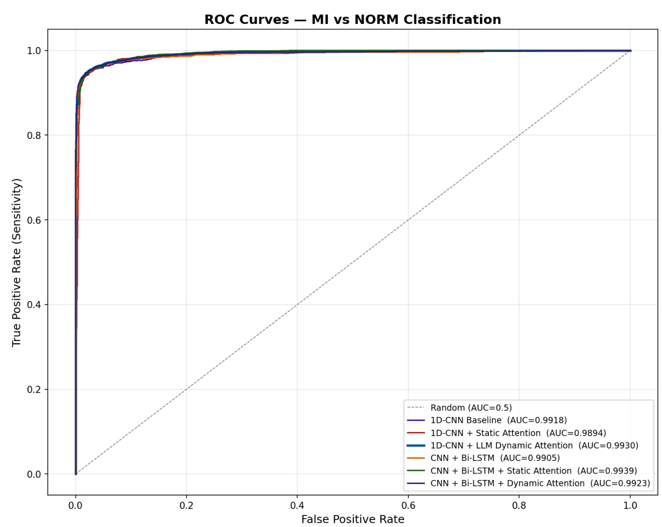

Fig. 10. Comparison of ROC-Curves for different Models

C. Conclusion

The study introduced a context-aware ECG Diagnosis framework that combines deep learning architectures with a Large Language Model(LLM) guided dynamic attention. The suggested method uses adaptive attention mechanism that dynamically modifies model focus based on patient specific contextual information while maintaining clinical safety by decoupling the LLM from raw signal processing, in compare to traditional approaches that rely on static attention without extracting weights based on the context of the patient's. The suggested 1D CNN with LLM guided dynamic attention obtains the most balanced and reliable performance among all analyzed models when it is performed on PTB-XL Dataset. When compared to both baseline and static attention models, it achieves the greatest Accuracy(0.9642), F1-Score(0.9635) and AUC(0.9930) showing enhanced power and better trade-off between Precision and Recall. While hybrid CNN-BiLSTM

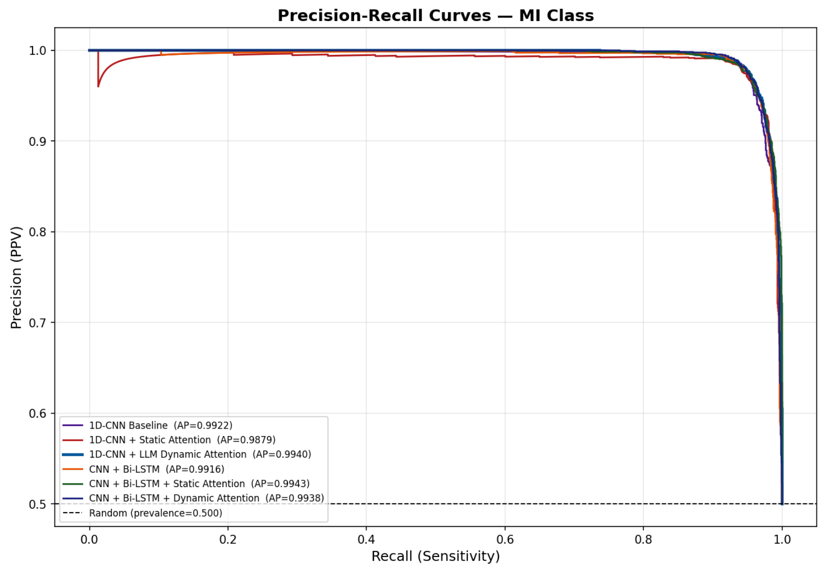

Fig. 11. Comparison of Precision Recall Curves for different Models

TABLE VIII
STATE OF THE ART QUANTITATIVE COMPARISON ON PTB-XL

Model	**Accuracy**	**F1-score**	**AUC**	**Task**
Rai et al. [20] 2022	0.962	0.962	0.991	MI vs NORM
Kumari et al. [10] 2025	0.956	0.957	0.994	ECG Classification
ECG-DAN [5] 2025	0.963	0.962	0.992	Risk Prediction
Kerdoudi et al. [9] 2026	0.961	0.960	0.992	ECG Classification
Proposed Model 2026	**0.964**	**0.963**	**0.993**	MI vs NORM

model captures temporal dependencies effectively, the results indicate that dynamic attention applied to a robust convolutional backbone is more effective in utilizing clinically significant information without introducing unnecessary model complexity. In addition to that while choosing diagnostically significant ECG leads for various patient profiles, the lead wise attention visualizations validate that the suggested model is consistent with clinical reasoning enhancing reliability and interpretability. This study also shows how LLM high level reasoning can be used as an attention controller instead of direct diagnostic agents improving models adaptability and reducing hallucinations. In conclusion, the suggested methodology provides a scalable and comprehensive solution for intelligent ECG analysis by effectively connecting the gap between static deep learning and context-aware clinical reasoning. In order to further evaluate the robustness of this study, future work will concentrate on expanding the model to multi-class cardiac diagnosis, adding deeper patient metadata and evaluating on real-world clinical situations.

REFERENCES

[1] Mohammed Yusuf Ansari, Mohammed Yaqoob, Mohammed Ishaq, Eduardo Feo Flushing, Iffa Afsa Changaai Mangalote, Sarada Prasad Dakua, Omar Aboumarzouk, Raffaella Righetti, and Marwa Qaraqe. A survey of transformers and large language models for ecg diagnosis: advances, challenges, and future directions. *Artificial Intelligence Review*, 58(9):261, 2025.

[2] Aram Avetisyan, Shahane Tigranyan, Ariana Asatryan, Olga Mashkova, Sergey Skorik, Vladislav Ananev, and Yury Markin. Deep neural networks generalization and fine-tuning for 12-lead ecg classification. *Biomedical Signal Processing and Control*, 93:106160, 2024.

[3] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*, 2014.

[4] Binqiang Chen, Yang Li, Xincheng Cao, Weifang Sun, and Wangpeng He. Removal of power line interference from ecg signals using adaptive notch filters of sharp resolution. *IEEE access*, 7:150667–150676, 2019.

[5] Chen Chen, Lei Li, Marcel Beetz, and Abhirup Banerjee. Large language model-informed ecg dual attention network. *IEEE Transactions on Big Data*, 2025.

[6] Chen Chen, Lei Li, Marcel Beetz, Abhirup Banerjee, Ramneek Gupta, and Vicente Grau. Large language model-informed ecg dual attention network for heart failure risk prediction. *IEEE transactions on big data*, 11(3):948–960, 2025.

[7] Mohammed Guhdar, Abdulhakeem O Mohammed, and Ramadhan J Mstafa. Advanced deep learning framework for ecg arrhythmia classification using 1d-cnn with attention mechanism. *Knowledge-Based Systems*, 315:113301, 2025.

[8] Saoueb Kerdoudi, Larbi Guezouli, and Tahar Dilekh. Enhanced ecg arrhythmia detection with deep learning and multi-head attention mechanism. *Computers and Electrical Engineering*, 131:110957, 2026.

[9] Saoueb Kerdoudi, Larbi Guezouli, and Tahar Dilekh. Enhanced ecg

arrhythmia detection with deep learning and multi-head attention mechanism. *Computers and Electrical Engineering*, 2026.
[10] KS Neelu Kumari, S Vijayabaskar, E Praynlin, and K Aravinda. Advanced ecg classification with improved optimized feature selection and attention cnn-bilstm technique. *Biomedical Signal Processing and Control*, 110:108299, 2025.
[11] Dan Li, Jianxin Zhang, Qiang Zhang, and Xiaopeng Wei. Classification of ecg signals based on 1d convolution neural network. In *IEEE Healthcom*, 2017.
[12] Dan Li, Jianxin Zhang, Qiang Zhang, and Xiaopeng Wei. Classification of ecg signals based on 1d convolution neural network. In *2017 IEEE 19th international conference on e-health networking, applications and services (Healthcom)*, pages 1–6. IEEE, 2017.
[13] Haijun Lin, Sirao Zhang, Qinghao Li, Ya Li, Jianmin Li, and Yuxiang Yang. A new method for heart rate prediction based on lstm-bilstm-att. *Measurement*, 207:112384, 2023.
[14] Yuxin Lin, Jing Ma, Wei Wang, Suyu Dong, Chaoyu Sun, Wanting Cong, Kuanquan Wang, and Gongning Luo. Transdiffecg: Semantically controllable ecg synthesis via transformer-based diffusion modeling. *Journal of Biomedical Informatics*, page 104948, 2025.
[15] Yue Ling. Bio+ clinical bert, bert base, and cnn performance comparison for predicting drug-review satisfaction. *arXiv preprint arXiv:2308.03782*, 2023.
[16] Ruoqi Liu, Yuelin Bai, Xiang Yue, and Ping Zhang. Teach multimodal llms to comprehend electrocardiographic images. *arXiv preprint arXiv:2410.19008*, 2024.
[17] Gouthamaan Manimaran, Abdolrahman Peimankar, Sadasivan Puthusserypady, Mahdi Momeni, Rifa Atul Izza Asyari, Masud Shah Jahan, Jonas Moll, Uffe Kock Wiil, and Ali Ebrahimi. Explainable deep learning based techniques for ecg-based heart disease classification: A systematic literature review and future direction. *Computers in Biology and Medicine*, 199:111324, 2025.
[18] Grace Ugochi Nneji, Md Altab Hossin, Kelvin Davies Mark, Edwin Sunday Umana, Goodness Temofe Mgbejime, and Jianping Li. Enhancing ecg classification in cardiac diagnostics: A novel approach using adaptive focal cross-entropy loss function. *IEEE Journal of Biomedical and Health Informatics*, 2025.
[19] Hari Mohan Rai and Kalyan Chatterjee. Hybrid cnn-lstm deep learning model and ensemble technique for automatic detection of myocardial infarction using big ecg data: Hybrid cnn-lstm deep learning model and ensemble technique for automatic detection of myocardial infarction using big ecg data. *Applied Intelligence*, 52(5):5366–5384, 2022.
[20] Hari Mohan Rai and Kalyan Chatterjee. Hybrid cnn-lstm deep learning model for automatic detection of myocardial infarction. *Applied Intelligence*, 2022.
[21] Paul Rubel, Danilo Pani, Alois Schloegl, Jocelyne Fayn, Fabio Badilini, Peter W Macfarlane, and Alpo Varri. Scp-ecg v3. 0: An enhanced standard communication protocol for computer-assisted electrocardiography. In *2016 Computing in Cardiology Conference (CinC)*, pages 309–312. IEEE, 2016.
[22] Abhiram Sharma, R Srivats, Utkarsh Mishra, et al. Spectronet-lstm: An interpretable deep learning approach to cardiac anomaly detection through heartbeat sound analysis. *Computers in Biology and Medicine*, 196:110774, 2025.
[23] Nils Strodthoff, Patrick Wagner, Tobias Schaeffter, and Wojciech Samek. Deep learning for ecg analysis: Benchmarks and insights from ptb-xl. *IEEE journal of biomedical and health informatics*, 25(5):1519–1528, 2020.
[24] P Wagner, N Strodthoff, R Bousseljot, W Samek, and T Schaeffter. Ptb-xl, a large publicly available electrocardiography dataset (version 1.0. 1), physionet (2020).
[25] Jinning Yang and Wen Shi. Diagecg: An llm-driven framework for diagnostic reasoning via discretized ecg tokenization. *arXiv preprint arXiv:2508.15338*, 2025.
[26] Han Yu, Peikun Guo, and Akane Sano. Ecg semantic integrator (esi): A foundation ecg model pretrained with llm-enhanced cardiological text. *arXiv preprint arXiv:2405.19366*, 2024.
[27] Yubao Zhao, Jiaju Kang, Tian Zhang, Puyu Han, and Tong Chen. Ecg-chat: A large ecg-language model for cardiac disease diagnosis. In *2025 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6. IEEE, 2025.